

PORTER COUNTY AP, IN, USA

WMO: 725327

Lat: 41.453N

Lon: 87.006W

Elev: 770

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04846

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	3.6	-8.5	3.6	0.0	-3.9	4.6	4.7	26.6	28.0	24.3	24.4	8.9	280	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	90.5	74.4	88.2	73.3	85.6	71.5	77.7	86.5	75.8	84.1	74.4	82.1	10.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	134.6	83.3	73.1	126.6	80.5	72.0	121.6	79.3	41.6	86.5	39.8	84.2	38.3	82.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.4	18.3		-8.4	94.7	6.2	3.6	-12.8	97.3	-16.5	99.5	-19.9	101.5	-24.4	104.1
			DB	-8.9	80.7	6.1	2.0	-13.3	82.2	-16.8	83.4	-20.2	84.5	-24.6	86.0
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	24.7	27.3	38.3	49.9	60.5	69.8	73.4	71.5	65.4	53.4	41.3	29.8	
	DBStd	19.42	11.74	11.15	10.83	9.41	9.08	6.84	5.72	5.09	7.52	9.08	9.47	10.94	
	HDD50	2936	785	639	392	117	14	0	0	0	1	67	291	630	
	HDD65	6109	1249	1056	831	461	201	31	6	8	86	378	712	1090	
	CDD50	3135	1	3	28	113	339	595	725	665	462	171	29	4	
	CDD65	835	0	0	2	7	62	176	266	208	97	17	0	0	
	CDH74	7380	0	0	20	93	609	1555	2417	1653	883	149	1	0	
	CDH80	2377	0	0	3	13	179	526	851	488	286	31	0	0	
Wind	WSAvg	8.3	10.2	9.7	9.4	10.0	8.5	7.2	6.1	5.5	6.2	8.0	9.2	9.6	
Precipitation	PrecAvg	39.1	1.9	1.6	2.9	4.0	3.7	4.2	3.9	3.7	4.2	3.1	3.2	2.9	
	PrecMax	48.4	4.2	4.5	5.6	8.1	6.7	9.6	11.3	10.2	10.3	6.9	8.4	6.6	
	PrecMin	25.4	0.3	0.2	0.6	0.5	1.3	0.7	1.0	0.5	0.0	0.5	1.4	0.6	
	PrecStd	5.9	1.1	1.0	1.3	1.8	1.5	2.0	2.1	2.2	2.5	1.5	1.7	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	62.8	76.7	81.7	89.4	92.7	94.6	91.2	91.5	84.0	70.5	61.2	
		MCWB	54.5	54.8	62.8	63.0	70.5	73.8	78.2	76.4	72.7	69.4	57.9	57.4	
	2%	DB	50.6	54.4	66.9	76.1	84.9	89.7	90.4	88.2	87.8	78.4	64.2	54.3	
		MCWB	46.8	50.1	57.2	60.7	67.5	72.3	75.8	74.3	71.8	64.8	55.3	50.5	
	5%	DB	45.1	47.1	61.1	72.2	81.1	86.0	87.9	85.2	82.4	73.0	60.3	48.8	
		MCWB	41.9	42.7	53.5	59.4	65.6	71.1	74.7	72.4	68.7	62.6	53.8	45.7	
	10%	DB	40.0	41.6	55.6	67.0	76.5	82.4	84.6	82.2	79.2	68.6	55.5	44.6	
		MCWB	37.4	37.8	49.2	55.6	64.3	69.5	72.5	71.6	67.0	60.7	49.9	41.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	57.3	63.4	68.0	73.9	78.0	80.6	79.1	76.1	71.8	61.2	57.8
			MCDB	57.8	60.4	76.2	76.8	84.8	88.6	90.7	87.6	87.7	81.9	66.1	60.6
2%		WB	47.8	50.2	59.4	63.8	70.7	75.3	78.5	76.7	73.7	68.2	58.0	51.7	
		MCDB	49.5	54.4	65.3	72.3	80.5	84.6	87.3	84.7	82.7	74.1	62.8	53.6	
5%		WB	41.8	43.6	54.9	60.5	68.3	73.5	76.6	74.9	71.5	64.8	54.3	45.9	
		MCDB	44.9	47.0	59.3	68.8	76.9	82.6	84.6	82.1	78.7	70.2	58.8	48.4	
10%		WB	37.5	37.9	49.7	57.5	66.2	71.7	74.9	73.2	69.3	61.7	50.6	41.5	
		MCDB	39.9	41.4	55.3	66.5	74.3	79.7	82.3	79.8	76.2	68.9	55.5	44.2	
Mean Daily Temperature Range	5% DB	MDBR	14.0	14.6	17.3	20.6	20.4	19.6	19.4	19.1	21.0	18.6	15.4	13.2	
		MCDBR	18.0	20.8	25.5	28.0	25.1	22.9	21.4	22.3	25.1	24.4	21.4	17.9	
		MCWBR	15.4	17.1	17.0	16.1	12.4	10.4	10.3	10.7	11.3	13.2	14.7	15.6	
	5% WB	MCDBR	17.5	20.1	23.1	25.1	21.4	19.7	19.1	18.6	20.5	20.2	18.7	16.6	
		MCWBR	15.7	17.6	16.7	16.4	12.1	10.4	10.3	10.2	10.4	13.1	14.7	15.5	
Clear-Sky Solar Irradiance	taub		0.298	0.324	0.345	0.400	0.429	0.437	0.453	0.447	0.420	0.356	0.319	0.297	
	taud		2.425	2.343	2.380	2.241	2.216	2.236	2.187	2.200	2.244	2.428	2.483	2.473	
	Ebn at Noon		272	279	286	276	270	267	261	259	257	265	261	262	
	Edh at Noon		25	31	34	42	44	44	45	43	39	29	23	22	
All-Sky Solar Radiation	RadAvg		509	777	1137	1470	1671	1918	1930	1681	1396	920	582	425	
	RadStd		50	100	88	123	150	120	98	98	115	95	70	44	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page